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Is internal carbon pricing an agent of change in the green transition?

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ABSTRACT

This study investigates the effect of internal carbon prices (ICP) on greenhouse gas (GHG) emission reductions. Using a sample of 57 firms over 13 years (2009–2021) with 741 observations, we find that firms adopting ICP significantly reduce their GHG emissions following the adoption. Specifically, following ICP adoption, firms reduced total GHG emissions by 2.677 times, Scope 1 emissions by 2.351 times and Scope 2 emissions by 0.326 times compared with their pre-2014 levels. Additionally, we find that the effectiveness of ICP on GHG emission reduction is enhanced through firms' commitment to renewable energy and carbon abatement initiatives. Our analysis also reveals that although participation in emissions trading systems (ETS) alone may not directly reduce emissions, ICP adoption within ETS-participating firms amplifies GHG reductions. Our study provides empirical evidence for the effectiveness of ICP in corporate climate management and its role in enhancing the impact of carbon market mechanisms. Overall, the findings suggest that an ICP can serve as an agent of change in corporate transition towards a net zero business environment.

1. Introduction

Climate change is increasingly attracting public and corporate attention, necessitating the transition towards a low-carbon economy. Such a transition has been driven by government support for stringent carbon legislation and regulations, including the Paris Agreement, the COP26 Glasgow Climate Pact and the United Kingdom's requirement for listed firms and financial institutions to disclose their plans for achieving net zero emissions, as well as the International Sustainability Standards Board's sustainability disclosure standards (Technical Readiness Working Group, 2021). Another key government policy to tackle climate change is the creation of an emissions trading system (ETS), whereby participants can trade carbon allowances and assets, creating a market-determined carbon price to encourage firms to reduce their carbon footprint (Luo et al., 2012). The use of an internal carbon price (ICP) is also gaining traction as a management tool to encourage decarbonisation within organisations. Involving pricing carbon emissions and raising awareness among employees, the ICP is a voluntary method for firms to take into account the implicit cost of carbon, even if they are not currently subject to external carbon regulations (World Business Council for Sustainable Development, 2015). The literature has reported a range of motivations for ICP adoption, including carbon urgency, carbon stringency, climate policy

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uncertainty and corporate governance (Bento & Gianfrate, 2020; Chen et al., 2023; Trinks et al., 2022).

To mitigate climate change concerns and facilitate decarbonisation, Stechemesser and Guenther (2012) claim that holistic carbon accounting is necessary to increase carbon transparency. This would allow management to comply with regulations and material flow and improve carbon efficiency (Velte et al., 2020). The extant literature further suggests that carbon accounting can facilitate an emissions-reduction process through costing (Tsai et al., 2012) and compliance with regulations (Zhang et al., 2020). For example, Tsai et al. (2012) suggest that although the conventional system of cost accounting is limited to precisely estimating the environmental costs related to carbon, an activity-based costing method can solve such concerns. In addition, a carbon value flow analysis for carbon emissions reduction introduced by Zhang et al. (2020) was specifically designed for the manufacturing sector when firms are subject to mandatory carbon regulations.

The accounting literature related to carbon emissions covers various mechanisms, including carbon disclosure, carbon performance, carbon management and carbon assurance (Clarkson et al., 2015; Datt et al., 2018; Luo et al., 2012; Tang & Luo, 2014). Although this literature has investigated various carbon accounting and management mechanisms, studies that have articulated the effect of an ICP on reducing greenhouse gas (GHG)[1] emissions remain exceedingly sparse. Among a few studies on ICP and its impact on GHG emission, Zhu et al. (2022) find that ICPs create corporate dynamic capability that leads to a lowered extent of environmental risk, such as carbon emissions. More precisely, Zhu et al. (2022) suggest that the process of carbon emission reduction by ICPs can be positively moderated by carbon dependence and corporate climate change targets. Building on literature, we revisit the effects of ICPs on GHG emission reduction; additionally, we delve into how an ICP facilitates the internalisation of carbon costs to accommodate an ETS to reduce GHG emissions.

Our first conjecture focuses on the direct relationship between ICP and GHG emission reduction. We posit that the adoption of an ICP facilitates GHG emission reduction through three underlying mechanisms. First, by directly implementing a carbon charge, an ICP assigns a financial cost per metric tonne of carbon dioxide equivalents emitted by a firm unit, thereby discouraging excessive GHG emissions from regular operations. Second, an ICP optimises resource allocation for environmentally friendly investments through shadow pricing, which assesses the risks associated with investments and business strategies in anticipation of future carbon regulations. Anecdotal evidence shows that shadow pricing helps firms align their ICP with market conditions, ensuring that their carbon risk exposure is minimised and investments in low-carbon technologies are prioritised. Third, an ICP enhances the estimation of the marginal abatement cost of carbon, allowing firms to determine the precise costs associated with reducing emissions, including expenses for renewable energy sources, green initiatives, carbon offsets and staff training in low-carbon technologies. To test our conjunctures, we adopt two empirical proxies of the mechanisms, namely renewable energy produced by firms and carbon abatement costs. We find that firms' substantive efforts towards an ICP—as evidenced by increased renewable energy and carbon abatement investments—tend to mitigate GHG emissions.

Our second conjuncture focuses on the nexus between an ICP and an ETS in GHG emission reduction. An ETS functions as a market-driven climate policy tool that sets a cap on GHG emissions, incentivising firms to adopt technological innovations and optimise production processes to reduce emissions. An ICP can complement an ETS by internalising the cost of carbon emissions in financial planning, promoting investments in low-carbon technologies and ensuring compliance with ETS regulations.

To empirically test the impact of an ICP on GHG emission reduction, we utilise a sample of 57 firms over a 13-year period from 2009 to 2021. According to the Carbon Disclosure Project (CDP) database, these firms all adopted an ICP in 2014 and have maintained continuous ICP adoption from that time. This resulted in a balanced panel dataset comprising 741 observations. Our findings suggest a negative relationship between the adoption of an ICP and firm GHG emissions, indicating an ICP has a positive effect on GHG emissions reduction. Our findings indicate that, on average, firms significantly reduced their total GHG emissions after the adoption of an ICP. Specifically, ICP-adopting firms produced 2.677 times less total GHG emissions (sum of Scope 1 and Scope 2 emissions), 2.351 times less Scope 1 GHG emissions and 0.326 times less Scope 2 GHG emissions than before 2014 and the adoption of an ICP. This finding is consistent with Zhu et al. (2022) who indicate a positive effect from ICPs on environmental performance. In the mechanism analysis, we find that the effectiveness of ICPs for reducing GHG emissions is strong in firms that make substantive efforts towards carbon neutrality. In particular, in firms that produce renewable energy and prioritise investment in carbon abatement, an ICP can significantly and positively moderate GHG emissions.

We also find that although participation in an ETS does not necessarily lead to carbon reduction, an ICP can strengthen the ETS in two ways. First, ICPs are more effective in firms participating in an ETS than in those that do not. Second, the results indicate that firms can adopt an ICP even without participating in an ETS, and that the adoption of an ICP can still lead to a reduction in GHG emissions—even without the direct influence of an ETS. Our empirical evidence shows that, on average, among ETS participants, firms that adopt an ICP generated significantly lower total GHG emissions compared with their emissions pre-adoption. Specifically, among ETS participants, after the adoption of an ICP, firms produced 3.485 times less total GHG emissions, 2.946 times less Scope 1 GHG emissions and 0.539 times less Scope 2 GHG emissions. Our results show a dynamic interactive relationship between ICPs and the external carbon price determined by an ETS, indicating that an ETS strengthens the effect of ICPs on carbon reduction. Our results are robust using the Heckman two-stage test, the two-stage least-square Lewbel's (2012) test and the Oster (2019) test for endogeneity.

This paper makes four incremental contributions to the literature in terms of corporate climate strategies and carbon performance. First, adopting transition management theory, we provide novel evidence as to the effectiveness of ICPs for reducing firms' absolute GHG emissions, including total emissions, Scope 1 and Scope 2 emissions. Although ICPs have gained traction as a managerial tool for

[1] In this paper, the terms GHG and carbon are used interchangeably.

driving climate action, systematic evidence on their real-world effectiveness has been limited. Building on transition management theory, Cooke (2010) outlines several transition steps—including structures for innovation systems and developing transformative processes—that can contribute to greater corporate sustainability. Using a unique research design, we extend the literature and document that ICPs can only attenuate GHG emissions post adoption given long-term commitments towards carbon neutrality. In the contexts of increasing environmental and global climate change problems, as well as GHG emissions management, we use an emissions-reduction facilitation hypothesis to extend the understanding of transition management in carbon accounting literature (Chen et al., 2023; Finger et al., 2005; Weber, 2003). Specifically, we provide evidence on the capacity of corporate transitional management-imposed ICPs to facilitate GHG emissions reduction. Our results are fresh evidence for the importance of nuanced sustainable transition studies, demonstrating that ICP adoption is associated with economically meaningful reductions in GHG emissions.

Second, we show that the presence of an external carbon-pricing regime—via an ETS, for example—amplifies the emission reduction effects of an ICP. This finding highlights the complementary relationship of internal and external carbon-pricing instruments and suggests that firms respond more aggressively to internal carbon costs when facing regulatory carbon constraints. That is, an ICP is viewed as an innovative tool that assigns a monetary value to GHG emissions for the purpose of managing carbon within an organisation. However, although researchers have studied the driving forces behind the adoption of an ICP (e.g., Bento & Gianfrate, 2020; Chen et al., 2023; Trinks et al., 2022), the outcomes for firms adopting ICPs during the decarbonisation process remain limited. Whereas firms under an ETS are highly motivated to adopt an ICP (Bento & Gianfrate, 2020; Chen et al., 2023; Trinks et al., 2022), we extend this literature by highlighting how an ICP can increase the effects of an ETS in reducing GHG emissions.

Third, we unpack the mechanisms through which ICPs drive decarbonisation. Specifically, we show that ICP adoption leads to greater investment in renewable energy generation and higher carbon abatement expenditures. These results suggest that ICPs help translate climate intentions into operational commitments by altering investment decisions and internal resource allocations.

Fourth, whereas Zhu et al. (2022) provide important early insights into the role of ICPs in enhancing corporate dynamic capabilities and improving environmental performance—primarily using firms from the United States of America (US) and focusing on emission intensity metrics—our study extends and complements their work in three important ways. First, we move beyond emission intensity to examine absolute reductions in GHG emissions, providing a direct and robust measure of firms' carbon mitigation outcomes. Second, we broaden the geographical scope to an international sample of 57 firms across 21 countries, allowing us to generalise the findings beyond the US context and account for cross-country heterogeneity in climate policy regimes. Third, our study introduces and empirically tests the interactions between ICPs and ETS, offering novel evidence as to how ICPs can enhance the effectiveness of external market-based carbon regulation. Additionally, whereas Zhu et al. (2022) theorise mechanisms such as innovation and green production, our study operationalises renewable energy generation and carbon abatement costs, offering more granular insights into how ICPs translate into carbon reductions through the spectrum of transition management. Thus, by addressing outcomes that are both proxied by absolute emission reductions and contextual via ETS interaction dimensions not previously captured in the ICP literature, our paper makes a distinct and incremental contribution.

The study is organised in the following way. Section 2 outlines the literature review and hypothesis development, Section 3 details data and the methodology and Section 4 presents the findings, analysis and robustness tests. Section 5 concludes the study.

2. Literature and hypotheses

2.1. Transition management theory

Organisations in many sectors transitioning to decarbonised business models. This process can be effectively explained through the framework of transition management theory, which views sectors not as simple technical arrangements but as complex socio-technical systems (Markard et al., 2012). These systems are characterised by an intricate web of networks, institutions and prevailing knowledge that jointly shape organisational and industrial behaviours and create momentum towards system transformation (Finger et al., 2005). A transition towards decarbonisation, therefore, is not a simple technological swap but a profound and multi-faceted transformation that entails concurrent shifts in organisational structures, political alignments, social practices and cultural values.

Within this complex landscape, mechanisms like ICPs serve as a critical catalyst for change. By assigning a financial cost to emissions, an ICP functions as a powerful institutional reform within a firm. It directly influences corporate strategy, reorients investment decisions and modifies behaviour across the organisation, thereby operationalising the broader goal of GHG reduction at a micro-level (Smith et al., 2005). Such systemic change requires active and intentional governance to ensure it drives sustainable outcomes. This is the core function of transition management. A goal-oriented approach to steering complex societal shifts, transition management involves providing clear direction, fostering collaboration between diverse stakeholders, evaluating influencing factors and monitoring the transition's progress (Loorbach & Rotmans, 2010).

Because the successful implementation of an internal corporate policy like an ICP is often interconnected with and supported by the wider institutional environment, favourable political and regulatory frameworks are also crucial for accelerating the transformation. Consequently, managing a decarbonisation transition requires a dual focus: implementing effective internal instruments while simultaneously engaging with the external system to build coalitions that support the shift to a low-carbon economy on a regional, national and global scale. Although transition management theory provides a highly relevant framework for analysing decarbonisation, its specific application in evaluating the role of ICPs within this process remains under-explored.

2.2. Internal carbon pricing and GHG emissions reduction

The 2005 introduction of the European Union ETS has led to ICPs gaining popularity within organisations addressing climate change. Ahluwalia (2017) posits that setting a carbon price and integrating the price into firms' business planning has several advantages, including the recognition of risks and opportunities related to climate change, the encouragement of positive climate actions and a decrease in GHG emissions. More specifically, a firm's ICP assigns financial costs per metric tonne of carbon dioxide equivalents released by a unit within that firm (Chen et al., 2023), thereby internalising the cost associated with carbon emissions within firm operations.

Extant literature investigates the determinants of ICPs. Employing transition management theory, Chen et al. (2023) find that firms with high transition commitment, tendency, pressure and environmental capacity are more likely to adopt an ICP, thereby facilitating corporate transition towards decarbonisation. They also suggest that firms show a greater commitment to introducing an ICP when there are strong incentives from the market or regulatory authorities. Additionally, firms with superior carbon reduction strategies, high carbon urgency and that are under strong public pressure are more inclined to adopt an ICP to maintain the resources they need from the community in which they operate, thereby facilitating their zero-carbon transition.

In a similar vein, Bento and Gianfrate (2020) investigate the drivers of ICPs from the perspectives of demand for carbon reduction and policy uncertainty. They suggest that the level of ICP adoption relates to macroeconomic, regulatory, industry and firm-specific characteristics, including the national climate policy, the country's development status, industry-level carbon engagement and corporate governance. Bento et al. (2021) confirm that national climate policies significantly influence firm-level ICP adoption. That is, firms in countries with explicit carbon-pricing policies tend to adopt higher ICPs.

Building on these studies, Chang (2017) finds that firms may adopt an ICP in anticipation of future regulatory measures, thereby mitigating financial risks associated with carbon emissions. This practice is often driven by sustainability commitments and the desire to demonstrate corporate responsibility. Trinks et al. (2022) find that ICP adoption is driven by external carbon constraints and by firms' exposure to formal carbon-pricing systems. In particular, although stringent national climate policies are an important mechanism driving firms to adopt an ICP, neither the size of countries or uncertainty about future carbon costs play a significant role.

Although the literature examines the drivers of ICP adoption, the evidence concerning the capacity of ICP adoption to mitigate carbon footprint remains limited. A case study by Klevtun and Nilsson (2021) shows that ICPs can effectively integrate environmental considerations into business operations, suggesting that ICP adoption encourages organisations to reduce their carbon footprint by influencing investment decisions and promoting energy efficiency measures. Their findings suggest that firms should develop tailored ICP strategies to align with their specific environmental goals and operational contexts. Ma and Kuo (2021) examine the impact of ICPs on firm financial performance, finding that, despite imposing initial costs, ICPs ultimately contribute to long-term financial performance by driving efficiency and mitigating regulatory risks. Hence, firms may view an ICP as an investment in long-term sustainability and financial resilience rather than as a short-term cost. Referencing the dynamic capability perspective, Zhu et al. (2022) suggest that ICPs affect corporate environmental strategy by allowing firms to increase their dynamic capacity for mitigating environmental risk. Because ICP adoption is driven by stakeholder expectations and societal legitimacy, Zhu et al. (2022) find that firms with ICP adoption are more likely to reduce carbon emissions. That is, as a form of internal carbon governance, ICP adoption can facilitate environmental performance improvement and competitive advantage.

Drawing upon the insights of prior literature, ICPs play a crucial role in carbon emissions reduction by providing a financial signal that integrates emission costs into business operations. That is, ICPs can incentivise organisations to adopt carbon mitigation practices, invest in low-carbon technologies and make strategic decisions that align with carbon reduction goals (Ma & Kuo, 2021). Notably, however, the effects of ICPs on GHG emissions are not without drawbacks, with the tension imposed by financial constraints able to hinder the implementation and effectiveness of GHG emission reduction initiatives. For example, Eccles et al. (2014) suggest that although firms with strong carbon practices often achieve better long-term financial performance, the initial costs associated with implementing these practices can be a barrier for firms with limited financial resources. Bui and De Villiers (2017) highlight that financial constraints significantly affect the ability of firms to invest in environmental management, resulting in limited access to capital and higher costs of green technologies. Consequently, although ICPs play a pivotal role in mitigating GHG emission, their effectiveness can be limited by the financial capacity of firms to bear initial costs, invest in low-carbon technologies and manage short-term financial impacts.

Against this backdrop, our study delves into the mechanisms by which the reduction of GHG emissions is advanced by ICPs through a pivotal shift toward decarbonisation, including carbon charges, shadow prices and the costs associated with carbon abatement. First, an ICP promotes the reduction of GHG emissions by directly implementing a carbon charge, which assigns a financial cost per metric tonne of carbon dioxide equivalents emitted by a unit within the firm, helping to discourage GHG emissions from regular business operations. Thus, GHG emissions are transparent and readily understandable, facilitating management's ability to develop ways to reduce emissions. As a result, the carbon charge imposed by an ICP may enable firms to establish dedicated funds for green investments to lower their GHG footprint. In turn, effective carbon abatement investments could help firms better mitigate their GHG emissions.

Second, ICPs optimise the allocation of resources for environmentally friendly investments through shadow pricing, with shadow price determined with reference to the current or projected price of carbon in the market. This mechanism evaluates the risks associated with investments and business strategies in anticipation of future carbon regulations, with shadow pricing used in feasibility research for a project as a representation of the projected cost to comply with carbon regulations (Addicott et al., 2019). Other factors, such as market conditions or technology advancements, can also be considered by managers in pricing carbon. For instance, if a firm's internal price is higher than the external market price, an ICP can help the firm reduce its potential exposure to carbon risk. As a result, managers who utilise shadow pricing may also take into account the indirect costs of carbon, such as the elimination of subsidies on fossil fuels from government, stricter restrictions on the use of carbon-intensive materials and technological advancements for reducing GHG emissions.

Third, ICPs enhance the estimation of the marginal abatement cost of carbon, allowing for a more precise determination of the cost associated with reducing carbon emissions, including expenses incurred to meet climate regulations. Such expenses may include the purchase of renewable energy sources, the implementation of green initiatives, the generation of carbon offsets and training for staff in the use of low-carbon technologies, equipment and materials. Unlike shadow pricing, which is determined retrospectively according to a firm's past efforts to reduce GHG emissions, the marginal abatement cost of carbon underlines the cost of reducing a firm's carbon consumption and indicates the associated cost of carbon regulation. Consequently, ICP adoption helps management gain insight into the cost of abatement or compliance with environmental regulations, helping firms evaluate future carbon reduction programs and further reducing their GHG emissions. We illustrate our conjecture as to how ICPs drive GHG emission reduction in Fig. 1.

Given this discussion, we posit that the adoption of an ICP facilitates GHG emission reduction through the effects of carbon charges, the use of shadow pricing and the marginal abatement cost of carbon. This gives rise to the following hypothesis:

H1. *Firms that have adopted an ICP reduce their GHG emissions.*

2.3. Internalising an external carbon price

An ETS serves as a market-driven climate policy tool designed to attain carbon emission targets, typically on a national or regional scale. Characterised as a cap-and-trade mechanism, the system establishes a limit on a country's GHG emissions, with corresponding allowances allocated to entities subject to the constraint. Following allocation, these emission allowances can be traded within the associated carbon market, reflecting the marginal abatement costs incurred by firms (Bebbington & Larrinaga-González, 2008). Existing literature generally contends that ETS can offer significant financial incentives to firms for the reduction of GHG emissions (Pinkse & Kolk, 2009), facilitating firms' internalising of emission costs that were previously externalised to society. The additional revenue generated from the ETS can aid in reducing carbon emissions and conserving energy in two important ways. First, an ETS market mechanism can incentivise firms to adopt technological innovations. That is, the pressure to comply with regulations and the financial benefits from selling emission allowances can drive firms to invest in technological advancements to enhance technical efficiency and, thereby, lower energy consumption and carbon emissions (Ren et al., 2020). Second, GHG emission reduction can be achieved by optimising the allocation of production factors. For instance, firms involved in an ETS are more likely to decrease the proportion of their regulated industrial activities by divesting businesses with high emission profiles (Hu et al., 2020).

ETS-participating firms are highly motivated to adopt an ICP because such a measure is an essential mechanism in firms' translation of GHG emissions reduction (Chen et al., 2023). That is, ICPs complement ETS by encouraging firms to internalise the cost of carbon emissions in their financial planning and decision-making processes. For example, Equinor ASA (Ticker: EQNR)—a petroleum refining firm headquartered in Norway—adopts an ICP-embedded risk management framework to both complement external carbon pricing and engage environmental innovation. According to its 2021 CDP response, an ICP helps Equinor quantify the financial impact of its carbon emissions, enabling better decision-making and investment in low-carbon solutions.[2] For instance, Equinor uses an ICP to evaluate the cost–benefit of mitigating actions (e.g., transitioning to renewable energy sources and developing technologies like carbon capture and storage). Its framework also includes the regular monitoring of regulatory developments, ensuring that Equinor's operations are aligned with external carbon-pricing mechanisms such as the ETS. Another example is Veolia Environment (Ticker: VIE FP), which is in the infrastructure sector based in France. According to its 2017 CDP response, Veolia set an internal price of carbon of up to 100 euros per tonne of CO_2 equivalent and embedded this into its long-term planning and business strategies. This has influenced decisions on country selection and the pricing of business activity distributions. Veolia stated that the adoption of an ICP was strongly endorsed by the firm's environmental committee to ensure compliance with ETS regulations. Veolia has further stated that ICP adoption has mitigated sensitivity to carbon price fluctuations through price indexing clauses in contracts, maintaining financial resilience.

Firms participating in ETS can also perform better under an ICP because they are better prepared for the costs associated with carbon emissions. That is, firms using an ICP are more capable of anticipating carbon costs and investing in low-carbon technologies, thereby achieving ETS compliance more cost-effectively (World Bank, 2024). ICPs are further considered to add value to corporate environmental strategy through innovation offsets like corporate innovations under the Porter's hypothesis (Zhu et al., 2022). Although an ICP is inherently a self-regulation measure that enjoys more flexibility than government regulation, the adoption and effectiveness of any carbon strategy is dependent on management discretion (Luo & Tang, 2021). Consequently, firms adopting an ICP are more likely to invest in low-carbon technologies and energy-efficient processes, which can lead to greater and more sustained emission reductions. For instance, Astellas (Ticker: 4503 JP), which is a pharmaceutical firm from Japan, suggested in their 2021 CDP response that an ICP can ensure that investments in energy efficiency and low-carbon technologies are financially viable, prioritising projects based on their cost-effectiveness in reducing GHG emissions. Astellas applied a standard ICP of 100,000 yen per tonne of CO_2 emissions reduced. Their report suggested that their ICP has helped determine the justification for capital investments across various business divisions, including pharmaceutical technology, drug discovery research and sales activities. To incorporate its ICP in its operational process, Astellas established an Environmental Health and Safety Committee, which secures an annual budget (around one

[2] The CDP response refers to the open-ended textual responses in the carbon management section (Questions 5.1a, 5.1b and 5.1c) and the internal carbon-pricing section (Question 11.3) of the CDP report in the corresponding year.

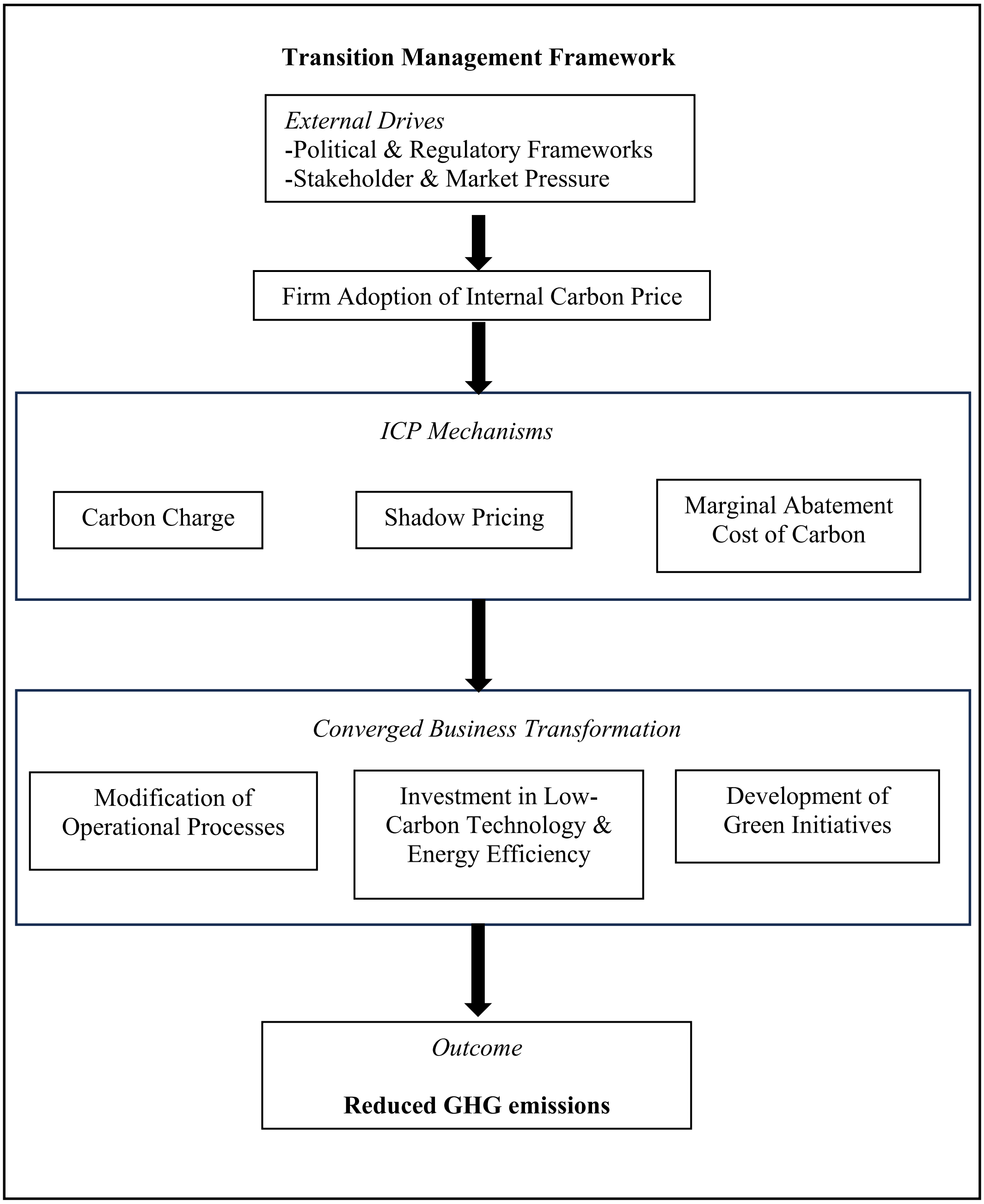

Fig. 1. Conceptual framework for GHG emission reduction via internal carbon pricing.

billion yen per annum) to support low-carbon initiatives. Astellas has further suggested that its ICP has been effective in accommodating external carbon costs, especially in board decision-making processes.

We posit that the adoption of an ICP strengthens the effects of external carbon pricing (i.e., an ETS) on a firm's GHG emissions. An ETS serves as a market mechanism to stimulate the reduction of GHG emissions, thereby formulating an external price for carbon. This market mechanism transfers carbon and climate change concerns into business operations and provides financial motivations for firms to transition to low-carbon operations (Ahluwalia, 2017). Specifically, it requires firms to incorporate the costs of their GHG emissions into their business bottom line and internalise the emission costs previously borne by society, creating strong economic incentives for firms to reduce their carbon emissions (Pinkse & Kolk, 2009). Given that the ETS is a direct reference for an ICP, we expect that firms that are participants in an ETS have a greater extent of engagement in internalising GHG emission costs. Thus, we expect that an ICP increases GHG emission reduction among firms participating in an ETS. This leads us to the second hypothesis:

H2. *The effect of an ETS on GHG emissions reduction is strengthened by an ICP.*

3. Research design

3.1. Sample and data

Given that CDP data began from 2009, our data spans from 2009 to 2021. Data about ICP adoption, GHG emissions (including *SCOPE*, *SCOPE1* and *SCOPE2*) and carbon management (including *ETS*, *CRI* and *CARIN*) were sourced from the CDP database. Financial and accounting data were collected from Bloomberg. Definitions of the variables are summarized in Appendix A.

Our sample selection process is guided by the conceptualisation of ICP to address our research question. As discussed in Section 2.1, an ICP is not a short-term initiative but a long-term strategic tool designed to be embedded within a firm's core operations (Ma & Kuo, 2021; Zhu et al., 2022). To accurately test ICP effectiveness, it is crucial to distinguish between firms with substantive, long-term ICP adoption and those whose adoption may be temporary, ceremonial or a failed implementation. Therefore, to ensure a clean and consistent treatment group for our panel data analysis, we include only firms with continuous ICP adoption. This selection allows us to better isolate the effects of a stable, institutionalised policy on GHG emissions reduction, comparing pre- and post-ICP adoption periods without the confounding variable of firm-level policy instability.[3] Consequently, our study is specifically designed to evaluate the environmental efficacy of ICPs that are successfully sustained over time. This approach strengthens the validity of our findings and provides a clearer understanding of how a committed ICP policy functions as a tool for decarbonisation.[4]

In our baseline models, we include firms that present continuous ICP adoption from 2014 to 2021, excluding those that discontinued ICP adoption. Given that ICP reporting was first required by the CDP in 2014, we manually assessed the pre-2014 carbon management responses (i.e., Questions 5.1a, 5.1b and 5.1c from the CDP) from our sample firms but found no mention of ICPs. Hence, we consider that ICPs were not adopted by the sample firms before 2014. This provides us a final number of 741 observations from 57 international firms between 2009 and 2021, whereby our sample firms are restricted to eight years of continuous ICP adoption. In our robustness analyses, we also included samples with multiple years of continuous ICP adoption (e.g., from 5 to 7 years), and our main findings remained consistent.

As shown in Table 1, the final sample consists of 741 observations across 21 countries and regions. In terms of sample distribution by country, as shown in Panel A, we note that the US contributes 15.8 %, followed by France (12.3 %), the United Kingdom (10.5 %) and Spain (10.5 %). Panel B indicates that the utilities, energy and industrials sectors contribute 22.8 %, 19.3 % and 17.5 %, respectively. Our sample distribution is overall consistent with literature that adopts data from the CDP (Chen et al., 2023; Luo et al., 2023; Shrestha et al., 2023).

3.2. Dependent variables

Following Luo and Tang (2016), *GHG emissions* are used to measure GHG emissions, which is a vector of (1) *SCOPE*, representing the sum of Scope 1 emissions and Scope 2 emissions, measured as (Scope 1 emissions + Scope 2 emissions)/1,000,000; (2) *SCOPE1*, representing the absolute emissions of Scope 1, measured as Scope 1 emissions/1,000,000; and (3) *SCOPE2*, representing the absolute emissions of Scope 2, measured as Scope 2 emissions/1,000,000. These measures are aligned with prior studies, which primarily focus on Scope 1 emissions and Scope 2 emissions (e.g., Chapple et al., 2013; Chen et al., 2023; Ioannou et al., 2016).[5] All variables are summarized in Appendix A.

3.3. Independent variables

The adoption of an ICP is used as the independent variable. In the baseline model, *ICP* is coded as 1 for post-ICP adoption in the eight consecutive years between 2014 and 2021, and it is coded as 0 for the firm before its adoption of an ICP (i.e., the years between 2009 and 2013). To examine the number of years from when firms adopted an ICP and a reduction in their GHG emissions was evident, we extend the *ICP* measure by introducing the alternative number of years of ICP adoption, namely *ICP5*, *ICP6* and *ICP7*, which we

[3] Our analysis follows the within-industry regression approach of Dyck et al. (2019), comparing pre- and post-ICP emissions within the treated group of continuous adopters. As a robustness check (see Panel A of Supplementary Table 1), we also analysed the full sample (including firms without ICPs) using an extended model based on Zhu et al. (2022) and found our results to be consistent.

[4] This structure of our data is viewed as a balanced panel dataset. The advantage of using panel data compared with a single cross-section or series of cross-sections with non-overlapping cross-section units is that it allows us to test and relax the assumptions that are implicit in cross-sectional analysis (Maddala, 2001). Compared with Zhu et al. (2022), our research design allows us to better compare the effects of ICPs on GHG emissions pre- and post-adoption across ICP firms and is less subject to the selection bias associated with the adoption of ICP.

[5] The rationale for focusing on the total amount of carbon emissions (i.e., absolute emissions) lies in its alignment with international climate science and policy frameworks, which aim to guide firms globally towards achieving net zero emissions. Absolute emissions provide a direct and transparent measure of a firm's environmental impact and are, therefore, more accurate for evaluating progress towards climate goals, such as net zero commitments and compliance with national or sectoral carbon budgets (IPCC, 2022). In contrast, carbon intensity metrics, such as emissions per unit of revenue or output, may decline through business growth rather than genuine decarbonisation. Importantly, global agreements such as the Paris Agreement explicitly prioritise absolute emission reductions as the foundation for climate stabilisation, with Article 4.4 of the Agreement urging developed countries to undertake economy-wide absolute emission reduction targets (UNFCCC, 2015). Nonetheless, in our supplementary analysis (see Panel A of Supplementary Table 1), our results are consistent using carbon emission intensity derived from an extended Zhu et al. (2022) model.

Table 1
Sample distribution.

Panel A: Sample distribution by country				
Country	# Observations	Observations%	#Firms with ICP adoption	ICP adoption%
Australia	26	3.5 %	16	3.5 %
Austria	26	3.5 %	16	3.5 %
Canada	13	1.8 %	8	1.8 %
Chile	13	1.8 %	8	1.8 %
Denmark	13	1.8 %	8	1.8 %
Finland	26	3.5 %	16	3.5 %
France	91	12.3 %	56	12.3 %
Germany	26	3.5 %	16	3.5 %
Hong Kong, China	13	1.8 %	8	1.8 %
Ireland	13	1.8 %	8	1.8 %
Italy	26	3.5 %	16	3.5 %
Japan	65	8.8 %	40	8.8 %
Netherlands	13	1.8 %	8	1.8 %
Norway	26	3.5 %	16	3.5 %
Portugal	26	3.5 %	16	3.5 %
South Africa	26	3.5 %	16	3.5 %
South Korea	13	1.8 %	8	1.8 %
Spain	78	10.5 %	48	10.5 %
Thailand	13	1.8 %	8	1.8 %
United Kingdom	78	10.5 %	48	10.5 %
United States of America	117	15.8 %	72	15.8 %
Total	741	100.0 %	456	100.0 %
Panel B: Sample breakdown by industry				
GIC Industry	# Observations	Observations%	#Firms with ICP adoption	ICP adoption%
Consumer Discretionary	78	10.5 %	48	10.5 %
Consumer Staples	78	10.5 %	48	10.5 %
Energy	143	19.3 %	88	19.3 %
Health Care	26	3.5 %	16	3.5 %
Industrials	130	17.5 %	80	17.5 %
Materials	52	7.0 %	32	7.0 %
Technology	39	5.3 %	24	5.3 %
Telecommunications	26	3.5 %	16	3.5 %
Utilities	169	22.8 %	104	22.8 %
Total	741	100.0 %	456	100.0 %

code 1 for firms' post-ICP adoption of five, six or seven years consecutively, respectively, and 0 for the firm before the adoption of the ICP.[6]

3.4. Moderators

An ETS ensures firms internalise emissions costs that are borne by society and generates strong financial incentives for firms to reduce GHG emissions. According to Pinkse and Kolk (2009), this system requires firms to take GHG emission reduction measures. In H2, we expect that firms that participate in an ETS are more engaged in the internalisation of GHG emissions costs, and that the effect of an ETS on GHG emissions reduction is strengthened by an ICP. Accordingly, to test H2, we code *ETS* as 1 for firms that are ETS participants, and 0 otherwise.

3.5. Model specification

To test H1 as to whether firms that have adopted an ICP reduce their GHG emissions, the following Eq. (1) is used:

$$\text{GHG Emissions} = \alpha + \beta_1\text{ICP} + \gamma_1\text{FSIZE} + \gamma_2\text{LEV} + \gamma_3\text{ROA} + \gamma_4\text{GROWTH} + + \gamma_5\text{TOBINSQ} + \gamma_6\text{ETS} + \gamma_7\text{CRI} + \gamma_8\text{CARIN} + \text{Industry FE} + \varepsilon_i \quad (1)$$

We then examine H2 as to whether firms that have adopted an ICP can further reduce their GHG emissions if they also participate in an external carbon price system via an ETS. The following Eq. (2) is used:

[6] The samples used for constructing *ICP*, *ICP7*, *ICP6* and *ICP5* consist exclusively of firms that have adopted an ICP for exactly and continuously 8, 7, 6 and 5 years, respectively. These samples are mutually exclusive and do not include observations from other subsets.

$$\text{GHG Emissions} = \alpha + \beta_1 \text{ICP} \times \text{ETS} + \beta_2 \text{ICP} + \beta_3 \text{ETS} + \gamma_1 \text{FSIZE} + \gamma_2 \text{LEV} + \gamma_3 \text{ROA} + \gamma_4 \text{GROWTH} + \gamma_5 \text{TOBINSQ} + \gamma_6 \text{CRI} + \gamma_7 \text{CARIN} + \text{Industry FE} + \varepsilon_i \tag{2}$$

In this study, a range of control variables are used. (1) Following Clarkson et al. (2008), larger firms tend to have higher GHG emissions because of their scale of operations. Therefore, we control for firm size (*FSIZE*), which is determined by taking the natural logarithm of the firm's total assets in US dollars. (2) Following Cormier and Magnan (2014), highly leveraged firms may face greater pressure to cut costs, which could influence their investment in emissions-reduction initiatives. We control for leverage (*LEV*), which is calculated by total liability divided by total assets. (3) More profitable firms have greater resources and incentives to invest in GHG emission reduction strategies, whereas less profitable firms may struggle to afford such investments. We, therefore, control for return on assets (*ROA*), which indicates the performance of the firm, calculated by dividing the profit by the total assets of a one-year lag (Al-Tuwaijri et al., 2004). (4) Economic growth influences changes in carbon emissions given it directly relates to energy consumption. To control for this, the growth variable (*GROWTH*) is created by subtracting one-year lagged sales from current sales and dividing the result by sales from a one-year lag (Cai et al., 2012). (5) Tobin's Q (*TOBINSQ*) represents market performance. Higher Tobin's Q can signal that the market values the firm's future growth and sustainability initiatives, potentially encouraging emissions reductions. We measure *TOBINSQ* as the sum of the total market capitalisation, the preferred shares, the total long-term debt and the current liability divided by total assets (Datt et al., 2018). (6) Firms that publicly declare their carbon reduction initiatives in reports such as the CDP are more likely to be proactive about reducing emissions. Carbon reduction incentives (*CRI*) are represented by a proxy and are coded as 1 if the firm addresses its carbon reduction initiatives in the CDP, and 0 otherwise (Chen et al., 2023). (7) The extent of GHG emission is subject to industry specifications. We control for carbon-intensive sectors (*CARIN*), which are represented by a binary variable. We code this as 1 if the firm is from the energy, materials or utility sectors, which are known to have a greater tendency towards transition, and 0 otherwise (Chen et al., 2023; Luo & Tang, 2014). We control for industry fixed effects to account for all industry-variant factors that might be jointly related to both our dependent variables and independent variables.[7]

4. Results and discussion

4.1. Descriptive statistics

Table 2 reports the descriptive statistics for the variables used in this study. As shown in Panel A, the mean and median values for *SCOPE* are 10.117 and 2.527, representing 10.117 and 2.527 million metric tonnes of GHGs per firm per year, respectively. Accordingly, the greatest contribution to *SCOPE* is sourcing from *SCOPE1*, with mean and median values of 9.004 and 1.719 million metric tonnes of GHGs. The emissions of *SCOPE2* are relatively small, with mean and median values of 1.114 and 0.538 million metric tonnes. This indicates a substantial right-skew across our GHG emission measures, with means exceeding medians, particularly for *SCOPE1*. On average, *SCOPE2* constitutes about 11 % (1.114/10.117) of total emissions, rising to roughly 21 % (0.538/2.527) at the median, suggesting that larger emitters are disproportionately Scope 1 dominant. The minimum of Scope 2 of zeros further points to a mass of low-to-negligible indirect emissions for a subset of firms. Also, there are 72.1 % firms participating in an ETS, suggesting that the majority of our samples are subject to explicit, market-based carbon-pricing constraints via cap-and-trade compliance obligations. The mean value of *ICP* (0.615), suggests that 61.5 % of our observations adopted an ICP consecutively between 2014 and 2021.

In terms of control variables, the average of *FSIZE*, *LEV*, *ROA*, *GROWTH* and *TOBINSQ* are 17.96, 0.318, 0.054, 0.427 and 1.770, respectively. Compared with Chen et al. (2023), our samples have high *FSIZE* and lower *LEV* and *ROA*, indicating that the sampled firms with eight consecutive years of ICP adoption are large firms. The mean of *CRI* suggests that 94.2 % of the firms sampled incorporate carbon control targets.

To analyse the differences among the variables pre- and post-ICP adoption, we provide descriptive statistics in Panel B of Table 2. We note that there are significant differences between *SCOPE*, *SCOPE1* and *SCOPE2* indicated by *t*-values, suggesting that GHG emissions after ICP adoption were reduced compared with pre-adoption levels.

4.2. Multicollinearity diagnosis

To prevent the possibility of extreme outliers, we applied winsorisation to all continuous variables at the 1st and 99th percentiles. Then, we carried out diagnostics for multicollinearity. The Pearson correlation is reported in Table 3, which indicates that all pairwise correlations between the independent and control variables are below 0.6. Additionally, we checked the variance inflation factor values and found that they were lower than the critical value of 10. Hence, we can confirm that there is no danger of multicollinearity.

[7] Given our sample selection—that is, all the firms in the model that have data available across 2009 to 2021—a standard year fixed-effects model is not applicable. This is because of the nature of our key variable of interest—namely, pre- and post-ICP adoption, *ICP*—whereby all the firms are coded as 1 after 2014, and 0 before 2014. Because of the time-invariant nature of *ICP* across the sampling periods (e.g., ICP was not adopted by any firms before 2014 but after 2014), such fixed effect can exacerbate measurement error problems and lead to biased estimates (Roberts & Whited, 2013).

Table 2
Descriptive statistics.

Panel A: Sample (2009–2021)

Variable	N	Min	Max	Mean	P50	SD	P25	P75
SCOPE	741	0.014	69.733	10.117	2.527	13.646	0.787	16.600
SCOPE1	741	0.006	69.082	9.004	1.719	13.056	0.240	15.273
SCOPE2	741	0.000	10.390	1.114	0.538	1.570	0.144	1.191
ICP	741	0.000	1.000	0.615	1.000	0.487	0.000	1.000
FSIZE	741	15.149	23.016	17.960	17.580	1.706	16.856	18.734
LEV	741	0.001	1.229	0.318	0.282	0.177	0.194	0.424
ROA	741	−0.237	0.772	0.054	0.039	0.075	0.023	0.068
GROWTH	741	−0.751	29.998	0.427	0.020	2.205	−0.061	0.119
TOBINQ	741	0.381	9.880	1.770	1.105	1.829	0.899	1.584
ETS	741	0.000	1.000	0.721	1.000	0.449	0.000	1.000
CRI	741	0.000	1.000	0.942	1.000	0.234	1.000	1.000
CARIN	741	0.000	1.000	0.667	1.000	0.472	0.000	1.000

Panel B: Sample (2009–2013; ICP = 0) versus sample (2014–2021; ICP = 1) [a]

	2009–2013; ICP = 0								2014–2021; ICP = 1								
Variable	N	Min	Max	Mean	P50	SD	P25	P75	N	Min	Max	Mean	P50	SD	P25	P75	*t*-value
SCOPE	285	0.038	67.790	11.462	2.527	16.023	0.777	17.800	456	0.014	69.733	9.276	2.602	11.864	0.796	15.625	−1.99**
SCOPE1	285	0.006	67.790	10.180	2.074	15.251	0.240	15.718	456	0.007	69.082	8.268	1.418	11.429	0.238	14.582	−1.82*
SCOPE2	285	0.000	10.390	1.283	0.559	1.846	0.157	1.280	456	0.000	7.680	1.008	0.537	1.361	0.131	1.096	−2.17**
FSIZE	285	15.139	22.713	17.838	17.565	1.674	16.833	18.657	456	15.317	23.016	18.037	17.581	1.723	16.881	18.917	1.56
LEV	285	0.001	0.708	0.285	0.256	0.154	0.174	0.383	456	0.001	1.229	0.339	0.299	0.187	0.208	0.438	4.30***
ROA	285	−0.096	0.772	0.051	0.038	0.067	0.026	0.067	456	−0.237	0.693	0.055	0.040	0.079	0.022	0.069	0.83
GROWTH	285	−0.411	1.250	0.053	0.039	1.168	−0.027	0.115	456	−0.751	29.998	0.661	0.006	2.783	−0.074	0.129	4.65***
TOBINQ	285	0.381	5.016	1.222	1.035	0.625	0.868	1.337	456	0.381	9.880	2.113	1.154	2.212	0.922	1.783	8.10***
ETS	285	0.000	1.000	0.642	1.000	0.480	0.000	1.000	456	0.000	1.000	0.770	1.000	0.422	1.000	1.000	3.69***
CRI	285	0.000	1.000	0.874	1.000	0.333	1.000	1.000	456	0.000	1.000	0.985	1.000	0.123	1.000	1.000	5.40***
CARIN	285	0.000	1.000	0.667	1.000	0.472	0.000	1.000	456	0.000	1.000	0.667	1.000	0.472	1.000	1.000	0.00

Note.
[a] *if *p* is significant at 10 % level; **if *p* is significant at 5 % level; ***if *p* is significant at 1 % level.

Table 3
Multicollinearity diagnostics: Pearson correlation matrix [a, b] (N = 741).

Variable	*ICP*	*FSIZE*	*LEV*	*ROA*	*GROWTH*	*TOBINQ*	*ETS*	*CRI*
ICP	1.000							
FSIZE	0.057	1.000						
LEV	0.150***	−0.273***	1.000					
ROA	0.029	0.038	0.115***	1.000				
GROWTH	0.134***	−0.004	0.364***	0.181***	1.000			
TOBINQ	0.237***	0.043	0.455***	0.594***	0.548***	1.000		
ETS	0.138***	0.133***	0.154***	−0.072*	0.043	−0.012	1.000	
CRI	0.231***	0.049	0.118**	0.015	0.042	0.080**	0.154***	1.000

Note.
[a] Variance inflation factor values are less than the critical value of 10, which are not tabulated but are available on request.
[b] * if *p* is significant at 10 % level; ** if *p* is significant at 5 % level; *** if *p* is significant at 1 % level.

4.3. Regression results for H1

In H1, we hypothesise that the adoption of an ICP can reduce firms' GHG emissions. Consistent with transition management theory, we find that the adoption of an ICP is associated with materially lower GHG emissions. The results in Table 4 show that the coefficients of *ICP* ($\beta1 = -2.677$, $t = -2.82$, $p < 0.01$; $\beta1 = -2.351$, $t = -2.60$, $p < 0.01$; $\beta1 = -0.326$, $t = -2.48$, $p < 0.05$) are negatively related to *SCOPE*, *SCOPE1* and *SCOPE2*, respectively. The results are not only statistically significant but also economically significant. Post-ICP adoption, firms produce 2.677 times less total GHG emissions, 2.351 times less Scope 1 GHG emissions, and 0.326 times less Scope 2 GHG emissions than their pre-ICP adoption before 2014. The larger absolute effect on Scope 1 is consistent with a direct carbon-charge channel disciplining internal processes, while the significant Scope 2 effect points to shadow pricing in investment screening and procurement (e.g., renewable energy sourcing). Taken together, the pattern indicates that ICPs coordinate abatement across interdependent routines inside firms, thereby supporting H1.[8]

For control variables, *ETS* and *CARIN* (except measured by *SCOPE2*) have positive impacts on GHG emissions. These findings are reasonable to explain why firms participate in an ETS and belong to carbon intensity industries (*CARIN*). This is because these firms can forecast their increases in GHG emissions in the future. We use the absolute GHG emissions as dependent variables—including *SCOPE*, *SCOPE1* and *SCOPE2*—and control with *GROWTH*, which reflects the intensity of firms' GHG emissions through their total production. The positive effects of *GROWTH* are only statistically significantly associated with *SCOPE* and *SCOPE1*. We find that this is because *SCOPE1* is the dominant contributor to total GHG emissions. Additional evidence supporting this argument can be found in Table 4, which shows the differences for the means and medians of *SCOPE1* and *SCOPE2*. *LEV* has a negative impact, while other firm-specific variables including *FSIZE*, *TOBINSQ*, *ROA* and *CRI* have no impact.

In the baseline model (Eq (1)), we measure *ICP* for the firms that have adopted an ICP for eight consecutive years relative to this figure, in the years prior to ICP adoption. We also assess the number of years following ICP adoption that it takes for firms to reduce their GHG emissions. To achieve this objective, we re-sample the data according to the numbers of years that firms adopted an ICP and proxy this as *ICP7*, *ICP6* and *ICP5*. These are coded as 1 if a firm adopted an ICP for seven, six or five consecutive years, respectively, or 0 prior to ICP adoption.

The results are shown in Table 5. In Columns (1)–(3), the coefficients for *ICP7* are ($\beta_1 = -2.936$, $t = -3.26$, $p < 0.01$; $\beta_1 = -2.425$, $t = -2.93$, $p < 0.01$; $\beta_1 = -0.511$, $t = -2.56$, $p < 0.05$) are negatively related to *SCOPE*, *SCOPE1* and *SCOPE2*, respectively. In Columns (4)–(6), the coefficients for *ICP6* are ($\beta_1 = -2.837$, $t = -3.16$, $p < 0.01$; $\beta_1 = -2.362$, $t = -2.89$, $p < 0.01$; $\beta_1 = -0.475$, $t = -2.27$, $p < 0.05$) are negatively related to *SCOPE*, *SCOPE1* and *SCOPE2*, respectively. In Columns (7)–(9), the coefficients for *ICP5* are ($\beta_1 = -2.601$, $t = -2.82$, $p < 0.01$; $\beta_1 = -2.126$, $t = -2.51$, $p < 0.05$; $\beta_1 = -0.475$, $t = -2.37$, $p < 0.05$) are negatively related to *SCOPE*, *SCOPE1* and *SCOPE2*, respectively. These findings not only support H1, but also suggest that (1) the more years of ICP adoption, the more GHD emissions were reduced; and (2) at least a consecutive five-year period of ICP adoption is required before firms begin reducing their GHG emissions.[9]

4.4. Regression results for H2

In H2, we predict that the effect of an ETS on the reduction of GHG emissions is strengthened by an ICP, given that firms that are participants of an ETS would be more engaged in the internalisation of GHG emissions costs. Using Eq (2) for this analysis, Table 6 suggests that the ETS participants in our sample are heavy emitters, as evidenced by the positive and significant coefficient estimates of

[8] To mitigate the likelihood of noise in firms affected by the Covid-19 pandemic, we remove the data for 2020 and 2021. The results (not tabulated in this paper) are consistent with the results presented in Table 4. In Panel B of Supplementary Table 1, we undertake a robustness analysis by incorporating two country-level variables (Worldwide Governance Indicators [*WGI*] and gross domestic product [*GDP*]) and country fixed effect. The results are consistent with the baseline results presented in Table 4.

[9] We also examine the samples of firms with a consecutive 4/3/2-year ICP adoption or the firms with discretionary ICP adoption between 2014 and 2021. The results (not tabulated in this paper) are not significant.

Table 4
Regression results for H1: The effects of internal carbon price on GHG emissions [a, b].

Variable		*SCOPE*	*SCOPE1*	*SCOPE2*
Intercept		(1)	(2)	(3)
	α	4.207	3.622	0.585
	t	(0.92)	(0.84)	(1.08)
Independent Variable:				
ICP	β_1	−2.677***	−2.351***	−0.326**
	t	(−2.82)	(−2.60)	(−2.48)
		–	[H1] –	–
Controls:				
FSIZE	γ_1	0.165	0.125	0.041
	t	(0.70)	(0.57)	(1.60)
LEV	γ_2	−11.071***	−10.165***	−0.906**
	t	(-3.66)	(−3.50)	(−2.45)
ROA	γ_3	−7.380	−8.735	1.355
	t	(−1.15)	(−1.39)	(0.94)
GROWTH	γ_4	0.945**	0.937**	0.008
	t	(2.08)	(2.06)	(0.25)
TOBINSQ	γ_5	−0.291	−0.299	0.008
	t	(−0.70)	(−0.74)	(0.14)
ETS	γ_6	4.163***	3.724***	0.439***
	t	(4.08)	(3.77)	(3.14)
CRI	γ_7	2.229	2.066	0.164
	t	(1.28)	(1.33)	(0.57)
CARIN	γ_8	7.899*	8.083*	−0.184
	t	(1.91)	(1.82)	(−0.37)
Industry FE		Yes	Yes	Yes
Adj. R^2		0.359	0.365	0.111
F-stat		29.52***	30.29***	11.13***
N		741	741	741

Table 4 reports the results of the effects of internal carbon price (ICP) on greenhouse gas (GHG) emissions. *SCOPE* represents the accumulated GHG emissions of *SCOPE1* and *SCOPE2*, measured as *SCOPE*/1,000,000. *SCOPE1* and *SCOPE2* are measured as *SCOPE1*/1,000,000 and *SCOPE2*/1,000,000, respectively. *ICP* is coded as 1 for post-ICP adoption at firm level between 2014 and 2021, and 0 for the years before the ICP adoption between 2009 and 2013.
Note.
[a] The first row (number) represents the estimated coefficient, the second row (number in parentheses) represents the *t*-value.
[b] * if *p* is significant at 10 % level; ** if *p* is significant at 5 % level; *** if *p* is significant at 1 % level.

ETS related to *SCOPE* ($\beta_3 = 6.134$, $t = 3.56$), *SCOPE1* ($\beta_3 = 5.39$, $t = 3.21$) and *SCOPE2* ($\beta_3 = 0.744$, $t = 3.6$). Importantly, our results show that the coefficient estimates of the interactive terms for *ICP* × *ETS* are negatively related to *SCOPE*, *SCOPE1* and *SCOPE2*. In terms of economic significance, on average, ICP adoption attenuates GHG emissions (*SCOPE*), Scope 1 emissions reduction (*SCOPE1*) and Scope 2 emissions reduction (*SCOPE2*) among ETS participants by 3.485, 2.946 and 0.539 times, benchmarking with non-ETS participants, respectively. This suggests that an ICP can strengthen an ETS in two ways. First, an ICP is more effective in firms participating in an ETS than in firms that do not participate in an ETS. Second, the results indicate that firms may adopt an ICP without participating in an ETS, and that the adoption of an ICP can still reduce GHG emissions, even without the direct influence of an ETS.[10] These findings show a dynamic interactive relationship between ICPs and external carbon prices determined by an ETS. That is, an ETS strengthens the effect of an ICP on carbon reduction, whereas an ICP can potentially substitute for an ETS in reducing emissions by firms that are excluded from an ETS. This insight has not been documented in the extant literature. Overall, the results support our hypothesis that ICPs play a critical role in carbon reduction. An ICP can either interact with ETS participants to reduce GHG emissions, or enable firms that do not participate in an ETS to reduce their emissions, which is in line with our argument. Thus, H2 is supported.

Next, we investigate the number of years required for the adoption of an ICP to strengthen the effect of an ETS in reducing GHG emissions. The results are shown in Table 7. In Columns (1)–(3), the coefficients for *ICP7* × *ETS* are ($\beta_1 = -5.176$, $t = -3.08$, $p < 0.01$; $\beta_1 = -3.982$, $t = -2.53$, $p < 0.05$; $\beta_1 = -1.194$, $t = -3.56$, $p < 0.01$) and are negatively related to *SCOPE*, *SCOPE1* and *SCOPE2*, respectively. In Columns (4)–(6), the coefficients for *ICP6* × *ETS* are ($\beta_1 = -4.518$, $t = -2.79$, $p < 0.01$; $\beta_1 = -3.396$, $t = -2.28$, $p < 0.05$; $\beta_1 = -1.122$, $t = -3.22$, $p < 0.01$) and are negatively related to *SCOPE*, *SCOPE1* and *SCOPE2*, respectively. In Columns (7)–(9), the coefficients for *ICP5* × *ETS* are ($\beta_1 = -2.831$, $t = -1.64$, $p > 0.1$; $\beta_1 = -1.761$, $t = -1.09$, $p > 0.1$; $\beta_1 = -1.07$, $t = -3.11$, $p < 0.01$) and are negatively related to *SCOPE*, *SCOPE1* and *SCOPE2*, respectively. These findings are consistent with the results reported in Table 6, suggesting that an ICP can enhance an ETS. Similarly, we find that this effect requires at least five consecutive years of ICP adoption.

[10] 14.17 % (105) of the observations did not participate in an ETS although they have adopted an ICP. The data of this study was NOT collected via a questionnaire survey, interviews, or an experiment.

Table 5
Regression results of the effects of ICP on GHG emissions measured by years of adoption [a, b].

Variable		*SCOPE*	*SCOPE1*	*SCOPE2*	*SCOPE*	*SCOPE1*	*SCOPE2*	*SCOPE*	*SCOPE1*	*SCOPE2*
Intercept		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	α	3.929	8.320**	−4.391***	3.562	7.809**	−4.246***	0.285	3.372	−3.087***
	t	(0.94)	(2.22)	(−4.72)	(0.84)	(2.11)	(−4.24)	(0.06)	(0.79)	(−3.28)
Independent Variable:										
ICP7	β_1	−2.936***	−2.425***	−0.511**						
	t	(−3.26)	(−2.93)	(−2.56)						
ICP6	β_1				−2.837***	−2.362***	−0.475**			
	t				(−3.16)	(−2.89)	(−2.27)			
ICP5	β_1							−2.601***	−2.126**	−0.475**
	t							(−2.82)	(−2.51)	(−2.37)
Controls:										
FSIZE	γ_1	0.247	0.018	0.228***	0.050	−0.123	0.173***	0.224	0.074	0.150***
	t	(1.12)	(0.09)	(5.28)	(0.25)	(−0.71)	(4.16)	(1.01)	(0.37)	(3.80)
LEV	γ_2	−14.514***	−13.940***	−0.573	−14.861***	−14.626***	−0.235	−14.495***	−13.822***	−0.673
	t	(−4.80)	(−4.87)	(−1.20)	(−4.74)	(−4.96)	(−0.47)	(−4.37)	(−4.43)	(−1.24)
ROA	γ_3	−20.793**	−24.337***	3.545*	−15.337*	−19.608***	4.271*	−21.612***	−20.758***	−0.854
	t	(−2.29)	(−3.06)	(1.68)	(−1.92)	(−2.78)	(1.94)	(−2.77)	(−3.36)	(−0.29)
GROWTH	γ_4	1.622	1.717	−0.094	−3.767	−3.768	0.001	−3.316	−3.192	−0.123
	t	(1.24)	(1.33)	(−0.74)	(−1.41)	(−1.58)	(0.00)	(−1.30)	(−1.40)	(−0.17)
TOBINSQ	γ_5	0.238	−0.321	0.558**	0.378	−0.008	0.386**	−0.170	−0.377	0.207
	t	(0.37)	(−0.56)	(2.57)	(0.77)	(−0.02)	(2.00)	(−0.30)	(−0.79)	(0.95)
ETS	γ_6	4.280***	3.784***	0.495**	5.814***	5.035***	0.780***	4.441***	3.558***	0.883***
	t	(4.02)	(3.85)	(2.17)	(5.41)	(5.17)	(3.03)	(3.71)	(3.19)	(3.59)
CRI	γ_7	0.019	−0.304	0.323	−0.355	−0.615	0.260	1.623	1.207	0.416*
	t	(0.01)	(−0.21)	(1.34)	(−0.23)	(−0.43)	(1.10)	(0.97)	(0.76)	(1.92)
CARIN	γ_8	12.512***	7.636***	4.876***	13.579***	7.956***	5.622***	12.616***	7.782***	4.834***
	t	(6.05)	(4.69)	(6.49)	(6.91)	(5.46)	(7.50)	(6.59)	(5.42)	(6.86)
Industry FE		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R^2		0.375	0.381	0.280	0.377	0.386	0.286	0.347	0.358	0.244
F-stat		41.46***	42.09***	9.97***	41.32***	41.11***	11.16***	36.35***	36.24***	10.16***
N		804	804	804	781	781	781	800	800	800

Table 5 reports the results of the effects of internal carbon price (ICP) on greenhouse gas (GHG) emissions. *SCOPE* represents the accumulated GHG emissions of *SCOPE1* and *SCOPE2*, measured as *SCOPE*/1,000,000; *SCOPE1* and *SCOPE2* are measured as *SCOPE1*/1,000,000 and *SCOPE2*/1,000,000, respectively. *ICP7*/*ICP6*/*ICP5* is coded as 1 for post-ICP adoption for 7/6/5 years consecutively, and 0 for the years before the ICP adoption.

Note.

[a] The estimated coefficient is shown from the first row; The *t*-value is shown from the second row in parentheses.

[b] * if *p* is significant at 10 % level; ** if *p* is significant at 5 % level; ***if *p* is significant at 1 % level.

Table 6
Regression results for H2: The effects of internal and external carbon prices on GHG emissions [a, b].

Variable		*SCOPE*	*SCOPE1*	*SCOPE2*
Intercept		(1)	(2)	(3)
	α	2.932	2.545	0.388
	t	(0.64)	(0.58)	(0.70)
Independent Variable:				
ICP × *ETS*	β_1	−3.485*	−2.946*	−0.539**
	t	(−1.93)	(−1.68)	(−2.21)
		–	[H2] –	–
ICP	β_2	−0.208	−0.264	0.056
	t	(−0.15)	(−0.19)	(0.29)
ETS	β_3	6.134***	5.390***	0.744***
	t	(3.56)	(3.21)	(3.60)
Controls:				
FSIZE	γ_1	0.184	0.140	0.044*
	t	(0.78)	(0.64)	(1.71)
LEV	γ_2	−11.469***	−10.501***	−0.968***
	t	(−3.79)	(−3.61)	(−2.60)
ROA	γ_3	−7.183	−8.569	1.386
	t	(−1.15)	(−1.39)	(0.99)
GROWTH	γ_4	0.955**	0.945**	0.010
	t	(2.10)	(2.07)	(0.31)
TOBINSQ	γ_5	−0.275	−0.285	0.010
	t	(−0.67)	(−0.71)	(0.19)
CRI	γ_6	1.859	1.752	0.106
	t	(1.06)	(1.11)	(0.37)
CARIN	γ_7	8.170*	8.312*	−0.142
	t	(1.88)	(1.79)	(−0.31)
Industry FE		Yes	Yes	Yes
Adj. R^2		0.362	0.367	0.115
F-stat		28.10***	29.09***	10.30***
N		741	741	741

Table 6 reports the results of the interactive effects of internal and external carbon prices on greenhouse gas (GHG) emissions. *SCOPE* represents the accumulated GHG emissions of *SCOPE1* and *SCOPE2*, measured as SCOPE/1,000,000; *SCOPE1* and *SCOPE2* are measured as *SCOPE1*/1,000,000 and *SCOPE2*/1,000,000, respectively. *ICP* is coded as 1 for post-ICP adoption at firm level between 2014 and 2021, and 0 for the years before the ICP adoption between 2009 and 2013. *ETS* represents the participation in an emission trading scheme (ETS), and is coded as 1 if a firm is a participant of ETS, otherwise, it is coded as 0.

Note.

[a] The estimated coefficient is shown from the first row; The *t*-value is shown from the second row in parentheses.

[b] * if *p* is significant at 10 % level; ** if *p* is significant at 5 % level; *** if *p* is significant at 1 % level.

4.5. *Mechanism analysis: Renewable energy and carbon abatement initiatives*

The implementation of an ICP plays a crucial role in advancing the reduction of GHG emissions through a strategic shift towards decarbonisation. As we discussed in the preceding section, this shift is driven by key mechanisms such as carbon charges, shadow pricing and the estimation of carbon abatement costs. Whereas a carbon charge directly assigns a financial cost per metric tonne of carbon dioxide equivalents emitted, creating transparency and incentivising firms to invest in green initiatives that lower their GHG footprint, shadow pricing optimises resource allocation by evaluating the risks associated with future carbon regulations, allowing firms to anticipate and mitigate potential carbon-related risks. According to He et al. (2022), investments in renewable energy and carbon abatement help firms develop unique resources and capabilities, such as innovations in energy efficiency and enhanced climate resilience, which can lead to a competitive advantage. Ultimately, these investments position firms for long-term sustainability and success in a low-carbon economy, aligning business practices with the growing global emphasis on environmental responsibility. Against this backdrop, this section specifically evaluates the effectiveness of ICPs in firms that demonstrate a long-term commitment to GHG emission reduction through renewable energy initiatives and carbon abatement efforts.

We surrogate the renewable energy produced (*REP*) as the natural logarithm of the renewable energy produced by the firm, and the carbon abatement initiatives (*CAI*) as the natural logarithm of total investment required over its lifetime for all carbon abatement initiatives of a firm that have ‘become fully operational/functional in realising CO_2 savings’ in the reporting year following (He et al., 2022). These proxies are incorporated as the interactions of *ICP* × *REP* (*CAI*) in the baseline model (Eq. (1)). As shown in Panel A of Table 8, the coefficients for *ICP* × *REP* are significantly and negatively related to *SCOPE*, *SCOPE1* and *SCOPE2*. Likewise, the coefficients shown in Panel B for *ICP* × *CAI* are significantly and negatively related to *SCOPE* and *SCOPE1*. These findings suggest that the effect of an ICP to reduce GHG emissions is stronger when firms have long-term commitments to reduce GHG emissions through their efforts to implement renewable energy and carbon abatement initiatives. These results are consistent with He et al. (2022), indicating the reduction of GHG emissions through renewable energy produced and carbon abatement initiatives.

Table 7
Regression results for the effects of internal and external carbon prices on GHG emissions measured by years of adoption [a, b].

Variable		*SCOPE*	*SCOPE1*	*SCOPE2*	*SCOPE*	*SCOPE1*	*SCOPE2*	*SCOPE*	*SCOPE1*	*SCOPE2*
		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Independent Variable:										
Intercept	α	−1.080	4.717	−5.797***	2.758	7.204**	−4.446***	−0.444	2.919	−3.363***
	t	(−0.24)	(1.20)	(−5.31)	(0.65)	(1.97)	(−4.33)	(−0.09)	(0.69)	(−3.48)
ICP7 × *ETS*	β_1	−5.176***	−3.982**	−1.194***						
	t	(−3.08)	(−2.53)	(−3.56)						
ICP7	β_2	0.679	0.356	0.323						
	t	(0.52)	(0.29)	(1.30)						
ETS	β_3	7.129***	5.976***	1.153***						
	t	(4.42)	(3.90)	(4.18)						
ICP6 × *ETS*	β_1				−4.518***	−3.396**	−1.122***			
	t				(−2.79)	(−2.28)	(−3.22)			
ICP6	β_2				0.193	−0.085	0.277			
	t				(0.16)	(−0.07)	(0.97)			
ETS	β_3				8.154***	6.793***	1.361***			
	t				(5.33)	(4.74)	(4.43)			
ICP5 × *ETS*	β_1							−2.831	−1.761	−1.070***
	t							(−1.64)	(−1.09)	(−3.11)
ICP5	β_2							−0.761	−0.982	0.221
	t							(−0.54)	(−0.75)	(0.80)
ETS	β_3							5.800***	4.404***	1.397***
	t							(3.68)	(2.97)	(4.69)
Controls:										
FSIZE	γ_1	0.261	0.030	0.232***	0.043	−0.128	0.171***	0.225	0.075	0.150***
	t	(1.20)	(0.15)	(5.38)	(0.22)	(−0.75)	(4.16)	(1.02)	(0.38)	(3.82)
LEV	γ_2	−14.879***	−14.221***	−0.658	−15.169***	−14.858***	−0.311	−14.585***	−13.878***	−0.707
	t	(−4.90)	(−4.95)	(−1.37)	(−4.82)	(−5.01)	(−0.62)	(−4.39)	(−4.44)	(−1.31)
ROA	γ_3	−19.667**	−23.471***	3.804*	−13.759*	−18.422***	4.663**	−20.453***	−20.037***	−0.416
	t	(−2.31)	(−3.08)	(1.92)	(−1.83)	(−2.72)	(2.21)	(−2.73)	(−3.29)	(−0.15)
GROWTH	γ_4	1.679	1.760	−0.081	−3.776	−3.775	−0.001	−3.242	−3.146	−0.095
	t	(1.28)	(1.36)	(−0.64)	(−1.43)	(−1.59)	(−0.00)	(−1.27)	(−1.38)	(−0.13)
TOBINSQ	γ_5	0.054	−0.462	0.516**	0.112	−0.208	0.320*	−0.296	−0.456	0.160
	t	(0.09)	(−0.83)	(2.45)	(0.24)	(−0.49)	(1.71)	(−0.53)	(−0.98)	(0.76)
CRI	γ_6	−0.375	−0.607	0.232	−0.620	−0.814	0.194	1.487	1.123	0.365*
	t	(−0.24)	(−0.42)	(0.99)	(−0.41)	(−0.57)	(0.83)	(0.88)	(0.70)	(1.68)
CARIN	γ_7	16.013***	10.079***	5.934***	13.514***	7.908***	5.606***	12.672***	7.817***	4.855***
	t	(7.71)	(6.57)	(7.10)	(6.85)	(5.38)	(7.51)	(6.61)	(5.43)	(6.89)
Industry FE		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R^2		0.381	0.385	0.289	0.382	0.389	0.293	0.348	0.358	0.251
F-stat		39.75***	40.52***	9.08***	39.71***	39.53***	10.21***	34.71***	34.50***	9.46***
N		804	804	804	781	781	781	800	800	800

Table 7 reports the results of the interactive effects of internal and external carbon prices on greenhouse gas (GHG) emissions. *SCOPE* represents the accumulated GHG emissions of *SCOPE1* and *SCOPE2*, measured as *SCOPE*/1,000,000; *SCOPE1* and *SCOPE2* are measured as *SCOPE1*/1,000,000 and *SCOPE2*/1,000,000, respectively. *ICP7/ICP6/ICP5* is coded as 1 for post-ICP adoption for 7/6/5 years consecutively, and is coded as 0 for the years before the ICP adoption. *ETS* represents participation in an emission trading scheme (ETS), and is coded as 1 if a firm is a participant of ETS; otherwise, it is coded as 0.

Note.

[a] The estimated coefficient is shown from the first row; The *t*-value is shown from the second row in parentheses.

[b] * if *p* is significant at 10 % level; ** if *p* is significant at 5 % level; *** if *p* is significant at 1 % level.

Table 8
Mechanism analysis [a, b].

Panel A: Renewable energy produced				
Variable		*SCOPE*	*SCOPE1*	*SCOPE2*
		(1)	(2)	(3)
Intercept	α	6.492	7.989	−1.497
	t	(0.43)	(0.56)	(−1.14)
ICP × REP	β_1	−0.791***	−0.680***	−0.111***
	t	(−3.06)	(−2.84)	(−3.06)
ICP	β_2	7.103**	5.926**	1.177***
	t	(2.39)	(2.16)	(3.05)
REP	β_3	1.841***	1.590***	0.251***
	t	(6.16)	(5.83)	(5.95)
FSIZE		−0.503	−0.541	0.038
		(−0.70)	(−0.79)	(0.73)
LEV		−13.621*	−15.021**	1.400**
		(−1.82)	(−2.08)	(2.18)
ROA		−9.672	−3.409	−6.263***
		(−0.54)	(−0.19)	(−3.69)
GROWTH		2.004***	2.042***	−0.038*
		(8.50)	(8.55)	(−1.79)
TOBINSQ		−0.850	−0.859	0.009
		(−1.30)	(−1.34)	(0.16)
ETS		−0.279	−1.157	0.878***
		(−0.09)	(−0.38)	(4.54)
CRI		0.706	1.890	−1.183**
		(0.17)	(0.49)	(−2.16)
CARIN		−10.530**	−10.957**	0.427
		(−2.13)	(−2.27)	(0.91)
Industry FE		Yes	Yes	Yes
Adj. R^2		0.383	0.383	0.365
N		286	286	286

Panel B: Carbon abatement initiatives				
Variable		*SCOPE*	*SCOPE1*	*SCOPE2*
		(1)	(2)	(3)
Intercept	α	−40.953***	−38.749***	−2.203**
	t	(−6.02)	(−6.09)	(−2.35)
ICP × CAI	β_1	−0.840**	−0.800**	−0.039
	t	(−2.40)	(−2.44)	(−0.80)
ICP	β_2	11.761**	11.372**	0.389
	t	(2.14)	(2.21)	(0.50)
CAI	β_3	1.646***	1.481***	0.166***
	t	(5.14)	(4.92)	(4.05)
FSIZE		1.297***	1.230***	0.067**
		(5.29)	(5.28)	(2.58)
LEV		−13.489***	−13.189***	−0.300
		(−4.22)	(−4.27)	(−0.92)
ROA		−1.203	−3.568	2.365**
		(−0.18)	(−0.51)	(1.98)
GROWTH		1.059**	1.081**	−0.022
		(2.40)	(2.44)	(−1.26)
TOBINSQ		−0.295	−0.281	−0.014
		(−0.71)	(−0.68)	(−0.31)
ETS		0.833	0.033	0.800***
		(0.69)	(0.03)	(7.11)
CRI		−1.941	−1.337	−0.604
		(−0.80)	(−0.61)	(−1.40)
CARIN		1.849	1.708	0.141
		(0.96)	(0.92)	(0.54)
Industry FE		Yes	Yes	Yes
Adj. R^2		0.467	0.476	0.211
N		624	624	624

Table 8 reports the results of the interactive effects of internal and external carbon prices on greenhouse gas (GHG) emissions. *SCOPE* represents the accumulated GHG emissions of *SCOPE1* and *SCOPE2*, measured as SCOPE/1,000,000; *SCOPE1* and *SCOPE2* are measured as *SCOPE1*/1,000,000 and *SCOPE2*/1,000,000, respectively. *ICP* is coded as 1 for post-ICP adoption at firm level between 2014 and 2021, and is coded as 0 for the years before the ICP adoption between 2009 and 2013. Renewable energy produced (*REP*) is determined by taking the natural logarithm of the renewable energy produced by the firm. Carbon abatement initiatives (*CAI*) are determined by taking the natural logarithm of total investment required over its lifetime

for all carbon abatement initiatives of a firm that have 'become fully operational/functional in realising CO_2 savings' in the reporting year.
Note.
[a] The estimated coefficient is shown from the first row; The *t*-value is shown from the second row in parentheses.
[b] * if *p* is significant at 10 % level; ** if *p* is significant at 5 % level; *** if *p* is significant at 1 % level.

4.6. Endogeneity tests

4.6.1. Heckman two-stage test

The decisions made by firms (e.g., self-selection of adoption) to adopt ICP might be endogenous and subject to various factors at both firm and country levels. Failing to account for these underlying factors that might influence firms' choices to adopt an ICP can lead to spurious estimates. Therefore, to eliminate sample self-selection bias, we run the Heckman two-stage test. As shown in Table 9, the dependent variable in the first-stage model is *RESICP*, which is coded as 1 if a firm responded to ICP-related questionnaires, and 0 otherwise. Three exogenous variables are used to construct the instrumental variable, namely the World Bank Voice and Accountability Index, the Yale Environmental Performance Index and the Climate Risk Index from Germanwatch. To mitigate the concern that these variables are likely to affect both the implementation of an ICP and GHG emissions, we follow Hasan et al. (2025) and undertake a strategy designed to effectively isolate the impact of country-level climate risk awareness. In particular, we regress emissions aggregated by sum at country level on the climate risk awareness variables as follows:

$$AWARENESS_{c,t} = \delta_0 + \delta_1 SCOPE_{c,t} + \varepsilon_{c,t}$$

Table 9
Heckman two-stage test [a], [b].

	(1)	(2)	(3)	(4)
	RESICP	*SCOPE*	*SCOPE1*	*SCOPE2*
IV1	0.001***			
	(3.20)			
IV2	0.001***			
	(6.94)			
IV3	−0.001***			
	(−8.38)			
FSIZE	0.117***	−0.599	−0.620	0.021
	(22.29)	(−1.20)	(−1.31)	(0.33)
LEV	0.411***	−11.813***	−11.354***	−0.458
	(5.94)	(−3.02)	(−3.05)	(−0.90)
ROA	−0.089	−9.711	−9.822	0.112
	(−0.54)	(−1.16)	(−1.23)	(0.10)
GROWTH	0.002	0.877***	0.904***	−0.027
	(0.73)	(3.74)	(4.05)	(−0.87)
TOBINSQ	0.084***	−0.911*	−0.892**	−0.019
	(9.85)	(−1.93)	(−1.99)	(−0.30)
ICP		−2.395***	−2.112**	−0.282**
		(−2.70)	(−2.50)	(−2.43)
IMR		−12.221**	−11.475**	−0.746
		(−2.34)	(−2.31)	(−1.09)
Intercept	−2.169***	36.398**	35.905**	0.492
	(−20.69)	(1.98)	(2.05)	(0.20)
Industry FE	Yes	Yes	Yes	Yes
Adj. R^2		0.360	0.366	0.149
F-stat		26.51***	27.13***	8.95***
N	14,767	726	726	726

Table 9 reports the results for the Heckman two-stage test. The dependent variable in the first-stage model is *RESICP*, which is coded as 1 if a firm responded to ICP-related questionnaires, and 0 otherwise. We adopted three exogenous variables to construct our instrumental variable, namely the World Bank Voice and Accountability Index, the Yale Environmental Performance Index and the Climate Risk Index from Germanwatch. To mitigate the concern that these variables are likely to affect both the implementation of internal carbon pricing and greenhouse gas emissions (GHG), we follow Hasan et al. (2025) and undertake a strategy designed to effectively isolate the impact of country-level climate risk awareness. In particular, we regress emissions aggregated by sum at country level on the climate risk awareness variables as follows:

$AWARENESS_{c,t} = \delta_0 + \delta_1 SCOPE_{c,t} + \varepsilon_{c,t}$

where *c* denotes countries and *t* denotes the year. The residuals from the equation, namely *IV1*, *IV2* and *IV3*, capture the aspect of country-level climate risk awareness not explained by GHG emissions according to the World Bank Voice and Accountability Index, the Yale Environmental Performance Index and the Climate Risk Index from Germanwatch, respectively. These residuals capture the variation in *SCOPE* that is orthogonal to each respective index and serve as valid instruments under the assumption that they are correlated with *SCOPE* but uncorrelated with the error term in the main regression. Note also that we developed instrumental variables according to Scope 1 or Scope 2 GHG emissions, and the results remain consistent when using instrumental variables constructed from total GHG emissions.

Note.
[a] The estimated coefficient is shown from the first row; The z/*t*-value is shown from the second row in parentheses.
[b] * if *p* is significant at 10 % level; ** if *p* is significant at 5 % level; *** if *p* is significant at 1 % level.

where *c* denotes countries and *t* denotes the year. The residuals from the equation—namely *IV1*, *IV2* and *IV3*—capture the aspect of country-level climate risk awareness not explained by GHG emissions according to the World Bank Voice and Accountability Index, the Yale Environmental Performance Index and the Climate Risk Index from Germanwatch, respectively. These residuals capture the variation in *SCOPE* that is orthogonal to each respective index and serve as valid instruments under the assumption that they are correlated with *Scope* but uncorrelated with the error term in the main regression. Note also that we developed instrumental variables based on Scope 1 or Scope 2 GHG emissions, and our results remain consistent with the baseline results when using instrumental variables constructed from total GHG emissions.

4.6.2. Two-stage least squares regression with Lewbel's (2012) approach

In our main analysis, we focused on the firms that continuously adopted ICP from 2014 to 2021, excluding those that discontinued its adoption. This approach limits our sample to firms with eight years of uninterrupted ICP adoption, allowing us to assess the effectiveness of ICPs in reducing GHG emissions before and after their implementation in firms with ICP adoption. Despite this sample selection approach, our results may still face endogeneity concerns, such as the potential omission of control variables. For instance, firms adopting an ICP may also implement other carbon management mechanisms and controls to mitigate emissions. To address this issue, we employ an instrumental variable approach following Lewbel (2012). In particular, Lewbel (2012) introduces a novel approach for identifying appropriate instruments in instrumental variable (IV) analysis, given that it can be challenging to find valid, relevant and strong instruments. The method proposed by Lewbel (2012) does not require external instruments, instead utilising the heterogeneity present in the error term of the first-stage regression to create instruments from within the model. This approach, according to Lewbel (2012), is useful in situations where external instruments are not available or are weak, and it can effectively address endogeneity issues. More specifically, this technique exploits model heteroskedasticity to generate instruments using available regressors, and the identification can be done by observing a vector of exogenous regressors that are not correlated with the covariance of heteroskedastic errors, which is a feature of many models in which error correlations are attributed to an unobserved common factor. Lewbel's (2012) technique has been widely used in recent studies to address endogeneity (e.g., Gong et al., 2018; Kao et al., 2018; Khoo & Cheung, 2022).

In the this study, we use Lewbel's (2012) approach to further validate the interpretation of our results by selecting from the generated instruments and applying the standard IV estimation method to the selected instruments. Table 10 presents the results of the

Table 10
Two-stage least squares (2SLS) with Lewbel's (2012) approach [a, b].

Variable	*SCOPE*	*SCOPE1*	*SCOPE2*
Intercept	(1)	(2)	(3)
	4.135	3.556	0.578
	(0.75)	(0.68)	(0.78)
Independent Variable:			
ICP	−5.56**	−4.993**	−0.570*
	(−2.43)	(−2.36)	(−2.09)
Controls:			
FSIZE	0.180	0.138	0.042
	(0.49)	(0.38)	(0.57)
LEV	−11.162	−10.248	−0.914
	(−1.48)	(−1.53)	(−0.81)
ROA	−10.424	−11.521	1.097
	(−0.712)	(−0.82)	(0.40)
GROWTH	0.920	0.914	0.006
	(1.48)	(1.45)	(0.18)
TOBINSQ	−0.022	−0.052	0.031
	(−0.03)	(−0.07)	(0.25)
ETS	4.505**	4.037*	0.468
	(2.41)	(1.91)	(1.27)
CRI	3.387	3.125	0.262
	(1.49)	(1.43)	(0.72)
CARIN	8.361	8.506	−0.145
	(1.44)	(1.46)	(−0.21)
Industry FE	Yes	Yes	Yes
Adj. R^2	0.050	0.049	0.020
Kleibergen Paap rk LM statistic	368.41***	368.41***	368.41***
Cragg–Donald F statistic	81.98***	81.98***	81.98***
N	741	741	741

Table 10 reports the results of the 2SLS Lewbel (2012) model to control for endogeneity. *SCOPE* represents the accumulated greenhouse gas emissions of *SCOPE1* and *SCOPE2*, measured as *SCOPE*/1,000,000; *SCOPE1* and *SCOPE2* are measured as *SCOPE1*/1,000,000 and *SCOPE2*/1,000,000, respectively. *ICP* is coded as 1 for post-ICP adoption at firm level between 2014 and 2021, and is coded as 0 for the years before ICP adoption between 2009 and 2013.

Note.
[a] The estimated coefficient is shown from the first row; The *t*-value is shown from the second row in parentheses.
[b] * if *p* is significant at 10 % level; ** if *p* is significant at 5 % level; *** if *p* is significant at 1 % level.

second-stage IV regression using Lewbel's (2012) approach. In the first-stage, in which the instruments are generated, we generate instruments using the control variables, consistent with our baseline regression. These instruments are selected because they are significant and important variables indicating GHG emission reduction. The results for the second-stage regression are reported in Table 10 for *SCOPE*, *SCOPE1* and *SCOPE2*, respectively, across Columns 1, 2 and 3. We find that the coefficients for the dependent variables remain negative and statistically significant, which corroborates the findings from the baseline analysis. With diagnostics, the under-identification test results (*LM* statistic) indicate that the excluded instrument is relevant; also, the Cragg–Donald Wald F-statistic exceeds the Stock and Yogo (2005) critical value, suggesting that our instrument does not suffer from weak identification.

4.6.3. Variable omission

Last, it is possible that our model omits factors that may co-move with both ICPs and GHG emissions. To mitigate these concerns, following Oster (2019), we conduct the omitted variable bias test as shown in Table 11. This test relies on the concept that creating an identifiable set is possible by considering the stability of coefficients along with R^2 obtained from regressions with and without controls. If the identifiable set does not include zero, it allows us to reject the null hypothesis that an omitted variable might be influencing the results. The identified set is calculated according to the following formula:

$$\left[\widetilde{\beta},\widetilde{\beta}-\delta\frac{(\beta_0-\widetilde{\beta})(RMAX-\widetilde{R})}{\widetilde{R}-R_0}\right]$$

where β_0 and R_0 denote the coefficient estimate and R^2 from the baseline model without control variables, respectively. $\widetilde{\beta}$ and $\widetilde{R}$ denote the coefficients estimates and R^2s from the baseline model with full set of controls, respectively. δ represents the relative significance of observed variables compared with unobserved variables in the generation of selection bias. Our main variable of interest is *ICP*, which is the independent variable, and our dependent variables are *SCOPE, SCOPE1* and *SCOPE2*. We present results according to Gao and Huang (2020) with $\delta = 1$ and RMAX = min (2.2 $\widetilde{R}$,1), and Mian and Sufi (2019) with $\delta = 1$ and RMAX = min (1.3 $\widetilde{R}$,1). We also use the extreme case of RMAX = 1 because R^2 cannot be greater than 100 %, following Oster (2019) and Zhang, Cui, Zheng, and Taylor (2025). According to these results, there is no evidence for omitted variables.

4.7. Cross-sectional analysis: Corporate financial constraints

In Section 2.1, we argue that financial constraints can hinder the implementation and effectiveness of GHG emission reduction initiatives. This is because the initial costs associated with implementing carbon practices can be a barrier for firms with limited financial resources (Eccles et al., 2014), limiting access to capital and resulting in higher costs of green technologies (Bui & De Villiers, 2017). Following these arguments, we examine the issue of financial constraints by dividing our sample using the median values of *KZ* (Kaplan & Zingales, 1997) and *WW* (Whited & Wu, 2006) indices. These results, shown in Panels A and B of Table 12, indicate that GHG emissions are reduced when firms have fewer financial constraints. This supports our conjuncture that the effect of an ICP on GHG emissions is greater when firms face fewer financial constraints, corroborating our main findings.

Table 11
Omitted variable bias tests.

Dependent Variables	*SCOPE*	*SCOPE1*	*SCOPE2*
Independent Variables	*ICP*	*ICP*	*ICP*
β_0	−2.185	−1.911	−0.275
R_0	0.006	0.005	0.007
$\widetilde{\beta}$	−2.677	−2.351	−0.326
$\widetilde{R}$	0.374	0.380	0.131
Oster Condition			
Assume $\delta = 1$; RMAX = min (1.3 $\widetilde{R}$,1)	[−2.827, −2.677]	[−2.485, −2.351]	[−0.342, −0.326]
Assume $\delta = 1$; RMAX = min (2.2 $\widetilde{R}$,1)	[−3.277, −2.677]	[−2.886, −2.351]	[−0.391, −0.326]
Assume $\delta = 1$; RMAX = 1	[−3.514, −2.677]	[−3.079, −2.351]	[−0.683, −0.326]

Table 11 presents the Oster (2019) bounds for our variable of interests as shown in the baseline regression in Table 4. The test relies on the concept that creating an identifiable set is possible by considering the stability of coefficients along with *R*-squares obtained from regressions with and without controls. If the identifiable set does not include zero, it allows us to reject the null hypothesis that an omitted variable might be influencing the results. The identified set is calculated according to the following formula:

$$\left[\widetilde{\beta},\widetilde{\beta}-\delta\frac{(\beta_0-\widetilde{\beta})(RMAX-\widetilde{R})}{\widetilde{R}-R_0}\right]$$

where β_0 and R_0 denote the coefficient estimate and R^2 from the baseline model without control variables, respectively. $\widetilde{\beta}$ and $\widetilde{R}$ denote the coefficients estimates and R^2s from the baseline model with full set of controls, respectively. δ represents the relative significance of observed variables compared with unobserved variables in the generation of selection bias. Our main variable of interest is *ICP*, which is the independent variables, and our dependent variables are *SCOPE, SCOPE1* and *SCOPE2*. We present results based on Gao and Huang (2020) with $\delta = 1$ and RMAX = min (2.2 $\widetilde{R}$,1), and Zhang et al. (2024) with $\delta = 1$ and RMAX = min (1.3 $\widetilde{R}$,1). We also use an extreme case of RMAX = 1 because, following Oster (2019) and Zhang et al. (2024), R^2 cannot be greater than 100 %.

Table 12
Corporate financial constraints [a, b].

Panel A: Measured by *KZ* Index

Variable	*SCOPE*	*SCOPE1*	*SCOPE2*	*SCOPE*	*SCOPE1*	*SCOPE2*
	KZ > Median			*KZ* ≤ Median		
	(1)	(2)	(3)	(4)	(5)	(6)
Intercept	2.931	2.261	0.670	−1.159	−2.536	1.377*
	(0.33)	(0.27)	(0.72)	(−0.19)	(−0.43)	(1.85)
ICP	−1.110	−0.845	−0.265	−3.550***	−3.201**	−0.349**
	(−0.85)	(−0.70)	(−1.26)	(−2.60)	(−2.44)	(−2.16)
FSIZE	0.495	0.452	0.043	0.059	0.080	−0.021
	(1.00)	(0.97)	(1.00)	(0.18)	(0.26)	(−0.65)
LEV	−22.652***	−22.127***	−0.525	6.582	7.393	−0.811
	(−4.35)	(−4.47)	(−0.79)	(1.09)	(1.28)	(−1.14)
ROA	−10.178	−12.147*	1.969	1.073	1.858	−0.786
	(−1.22)	(−1.72)	(0.81)	(0.12)	(0.21)	(−0.68)
GROWTH	1.013***	0.820***	0.193*	0.924	0.935	−0.012
	(3.69)	(3.32)	(1.97)	(1.44)	(1.47)	(−0.66)
TOBINSQ	−0.114	0.060	−0.174*	−1.217**	−1.290**	0.072
	(−0.24)	(0.14)	(−1.92)	(−2.11)	(−2.33)	(1.18)
ETS	1.621	0.482	1.139***	5.931***	5.852***	0.079
	(0.77)	(0.24)	(5.13)	(4.70)	(4.90)	(0.36)
CRI	−0.552	−0.171	−0.380	4.474**	3.525	0.950***
	(−0.17)	(−0.06)	(−0.73)	(1.98)	(1.63)	(3.76)
CARIN	7.273**	7.058**	0.215	11.222***	12.256***	−1.034*
	(2.47)	(2.56)	(0.49)	(4.53)	(4.89)	(−1.90)
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R^2	0.375	0.397	0.117	0.454	0.455	0.317
N	351	351	351	351	351	351

Panel B: Measured by *WW* Index

Variable	*SCOPE*	*SCOPE1*	*SCOPE2*	*SCOPE*	*SCOPE1*	*SCOPE2*
	WW > Median			*WW* ≤ Median		
	(1)	(2)	(3)	(4)	(5)	(6)
Intercept	−133.007***	−116.137***	−16.870***	45.727***	37.219***	8.508***
	(−9.55)	(−9.04)	(−6.71)	(6.03)	(5.26)	(6.41)
ICP	−1.730*	−1.495	−0.235	−3.735***	−3.365***	−0.370**
	(−1.76)	(−1.60)	(−1.42)	(−2.88)	(−2.71)	(−2.10)
FSIZE	7.922***	6.885***	1.037***	−2.748***	−2.369***	−0.379***
	(9.48)	(8.93)	(6.78)	(−7.57)	(−7.03)	(−5.98)
LEV	−9.244***	−8.839***	−0.405	2.206	2.320	−0.114
	(−2.67)	(−2.59)	(−0.77)	(0.45)	(0.49)	(−0.13)
ROA	−6.988	−7.110	0.122	0.619	0.047	0.571
	(−1.07)	(−1.17)	(0.06)	(0.04)	(0.00)	(0.35)
GROWTH	0.513***	0.497***	0.016	1.077**	1.097**	−0.019
	(3.56)	(3.68)	(0.70)	(2.55)	(2.52)	(−0.39)
TOBINSQ	−0.345	−0.290	−0.055	−0.948*	−0.985*	0.036
	(−0.87)	(−0.76)	(−0.71)	(−1.73)	(−1.84)	(0.57)
ETS	−5.052***	−5.214***	0.162	10.197***	9.920***	0.277
	(−3.21)	(−3.25)	(0.93)	(7.67)	(7.75)	(1.16)
CRI	1.383	1.650	−0.267	2.318	1.952	0.366
	(0.51)	(0.69)	(−0.57)	(0.97)	(0.87)	(1.16)
CARIN	14.356***	13.592***	0.764	2.255	1.781	0.475
	(8.06)	(7.45)	(1.63)	(0.88)	(0.73)	(1.44)
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R^2	0.570	0.566	0.211	0.565	0.566	0.301
N	351	351	351	351	351	351

Table 12 reports the results of the interactive effects of internal and external carbon prices on greenhouse gas (GHG) emissions. *SCOPE* represents the accumulated GHG emissions of *SCOPE1* and *SCOPE2*, measured as SCOPE/1,000,000; *SCOPE1* and *SCOPE2* are measured as *SCOPE1*/1,000,000 and *SCOPE2*/1,000,000, respectively. *ICP* is coded as 1 for post-ICP adoption at firm level between 2014 and 2021, and is coded as 0 for the years before ICP adoption between 2009 and 2013. *KZ* is the index–measurement of corporate financial constraints (Kaplan & Zingales, 1997); *WW* is the index–measurement of corporate financial constraints (Whited & Wu, 2006).
Note.
[a] The estimated coefficient is shown from the first row; The *t*-value is shown from the second row in parentheses.
[b] * if *p* is significant at 10 % level; ** if *p* is significant at 5 % level; *** if *p* is significant at 1 % level.

5. Conclusion

This study empirically examines the impact of ICPs on GHG emissions reduction and the role that ICPs play to internalise the external costs of carbon as induced by an ETS. According to evidence from 741 observations across 57 international firms over a 13-year period from 2009 to 2021, we find that the adoption of an ICP leads to a reduction in GHG emissions. Next, we find that the effect of an ETS with reference to reducing GHG emissions is strengthened by an ICP in two ways. First, an ICP is more effective in firms participating in an ETS than in firms that do not participate in an ETS. Second, the results indicate that firms may adopt an ICP without participating in an ETS, and the adoption of an ICP can still reduce GHG emissions without the direct influence of an ETS. These results remain consistent with alternative measures of ICP adoption and GHG emissions, indicating that decarbonisation did not occur until after at least five consecutive years of ICP adoption. Our mechanism analysis reveals that the effectiveness of an ICP in reducing GHG emissions is particularly strong in firms making substantial efforts toward carbon neutrality. Specifically, firms that actively produce renewable energy and invest in carbon abatement can significantly and positively enhance the impact of an ICP on reducing GHG emissions. In our cross-sectional analysis, we find that the effects of ICPs depend on the level of financial constraints faced by a firm. Our results remain robust via the Heckman two-stage test, the two-stage least squares test proposed by Lewbel (2012), and Oster (2019) for endogeneity check.

Our findings offer practical implications for regulators, policymakers, corporate board directors, investors and other stakeholders. First, our findings have implications for standard setters and policy makers concerned with the implementation of the International Financial Reporting Standards (IFRS) and the Final Rules by the US Securities and Exchange Commission for US-listed firms concerning sustainability and climate disclosure. Given that many corporations are facing mandatory disclosing requirements, integrating carbon cost internalisation into corporate strategies and risk management becomes critical for tackling GHG emission and climate change risks. This is crucial for compliance with the new standards.

Second, our findings are important to firm executives developing long-term GHG emission reduction or decarbonisation strategies. We document the real effects of ICPs in GHG emission reduction, showing that such effects are prominent in the long-run. This supports managers in ensuring and strengthening long-term resource allocation for decarbonisation strategies by indicating that such strategies may only be effective after a long-term commitment to carbon reduction. Third, our results are relevant for policymakers and regulators concerned with external carbon pricing. We document that internalising carbon costs with an ICP can effectively complement external carbon pricing through market mechanisms, thereby lowering GHG emissions. This is because ICPs can increase internal carbon transparency and lower carbon risks in operations.

Next, our findings suggest that ICPs inform strategic decision-making by highlighting the financial benefits of low-carbon investments, aiding firms to integrate these insights into their financial and operational strategies. This aligns with emphasis on governance and risk management disclosures from IFRS S2. By adopting an ICP, firms can better manage climate-related risks and improve their financial resilience, thus supporting their long-term sustainability and competitive advantage. Because of the enhanced carbon-pricing strategy induced by an ICP, firms that can more effectively engage stakeholders through comprehensive reporting on carbon-pricing strategies can enhance corporate reputation and stakeholder trust, fulfilling IFRS S1's broader disclosure requirements. Finally, our results are useful for investors and portfolio managers in the capital market. Our findings suggest that an ICP is a critical factor to consider in the evaluation of a firm's portfolio related to GHG emissions.

However, our study is not without limitations. The main independent variable, *ICP*, is measured as a dichotomous indicator according to pre- and post-ICP adoption. As a result, the extent of ICP implementation at the firm level is not observable in our analysis. Future studies should focus on examining the depth and scale of ICP implementation. Nonetheless, our findings suggest that a consistent carbon management strategy and a sustained commitment to carbon reduction are vital for achieving decarbonisation.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.bar.2025.101776.

Appendix A. Variable Definitions

Variable	Definition	Hypothesis	Sign	Source
Dependent variables				
SCOPE	The accumulated greenhouse gas (GHG) emissions of Scope 1 and Scope 2, measured as *SCOPE*/1,000,000.			CDP
SCOPE1	Scope 1 GHG emissions, measured as *SCOPE1*/1,000,000.			CDP
SCOPE2	Scope 2 GHG emissions, measured as *SCOPE2*/1,000,000.			CDP

(continued on next page)

(*continued*)

Variable	Definition	Hypothesis	Sign	Source
Independent variables				
ICP	The adoption of internal carbon price is coded as 1 if a firm adopted ICP for 8 years consecutively between 2014 and 2021, and 0 for the firm before the ICP adoption.	H1	–	CDP
ICP7	The adoption of internal carbon price is coded as 1 if a firm adopted ICP for 7 years consecutively, and 0 for the firm before the ICP adoption between 2009 and 2013.			CDP
ICP6	The adoption of internal carbon price is coded as 1 if a firm adopted ICP for 6 years consecutively, and 0 for the firm before the ICP adoption between 2009 and 2013.			CDP
ICP5	The adoption of internal carbon price is coded as 1 if a firm adopted ICP for 5 years consecutively, and 0 for the firm before the ICP adoption between 2009 and 2013.			CDP
Moderators				
ETS	The participation of a firm in an emission trading scheme, which reflects the external carbon price, is coded as 1 if the firm participates in the scheme, and 0 if it does not.	H2	–	CDP
REP	Renewable energy produced (*REP*) is determined by taking the natural logarithm of the renewable energy produced by the firm.			ASSET4
CAI	Carbon abatement initiatives (*CAI*) are determined by taking the natural logarithm of total investment required over the lifetime for all carbon abatement initiatives of a firm that have 'become fully operational/ functional in realising CO_2 savings' in the reporting year.			CDP
Other variables				
FSIZE	Firm size is determined by taking the natural logarithm of the total assets in US dollars.			Bloomberg
LEV	Leverage ratio is determined by taking total liability divided by total assets.			Bloomberg
ROA	Return on assets, calculated as the ratio of net profit to the previous period's total assets.			Bloomberg
GROWTH	Firm growth, computed as the difference between current sales and the sales from the previous year divided by the sales from the previous year.			Bloomberg
TOBINSQ	Tobin's Q, calculated by summing the total market capitalisation, the preferred shares, the long-term debt and the current liabilities and dividing by total assets.			Bloomberg
CRI	Carbon reduction incentive is indicated by a value of 1 if the firm responded to the CDP questionnaire with information about carbon reduction incentives; otherwise, it is coded 0.			CDP
CARIN	Carbon-intensive sector, coded as 1 if a firm is part of a sector that typically emits high levels of carbon, such as the energy, material or utility industry, and 0 otherwise.			CDP
CDPP	CDP participation, which is coded as 1 if a firm received the invitation of the CDP and responded 'yes' to participate, otherwise we code 0 if a firm declines to participate the CDP.			CDP
KZ	KZ index, measurement of corporate financial constraints (Kaplan & Zingales, 1997).			DataStream
WW	WW index, measurement of corporate financial constraints (Whited & Wu, 2006).			DataStream

Data availability

The authors do not have permission to share data.

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